
A3 Constructa

How the system works — a plain-language guide

A construction ERP built on ERPNext. This guide explains what the system does, how work flows through it, and where your money is tracked — without assuming you have used ERPNext before.

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Prepared for client walkthrough

1. The problem this solves

On a construction job, money is committed long before it is spent, and spent long before anyone sees a number.

You order steel in March. It arrives in May. It is used in June. The invoice is paid in July. By the time the cost appears in the accounts, the decision that caused it is four months old. Meanwhile the site is asking for more material, the client wants a progress claim, and nobody can say with confidence whether the substructure package is still inside its budget.

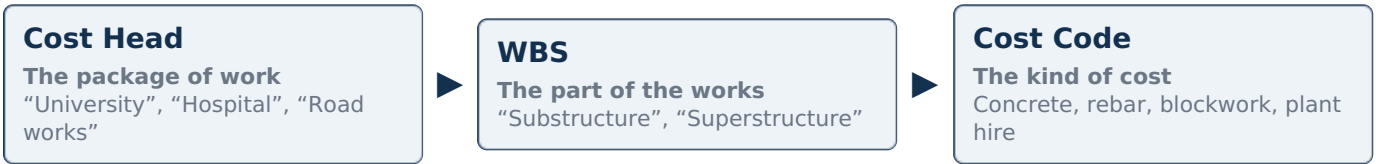
A3 Constructa closes that gap. Every purchase order, delivery, material issue and invoice is tagged to a part of the works, so at any moment you can answer three questions:

Question	Where the answer comes from
What did we budget?	The approved BOQ, allocated across the works
What have we committed?	Purchase orders raised but not yet invoiced
What have we actually spent?	Invoices posted and material issued from store

The difference between the second and third answer is the one most systems miss. A package can look healthy on spend and already be over-committed on orders that have not yet been invoiced. This system shows both side by side.

The three tags that make it work

Nearly everything in the system carries the same three tags. Once you understand these, the rest follows.



Every order, delivery, issue and invoice carries these three tags
That is what lets the system add up cost by package, by part of the works, or by type of cost.

In practice: when a storeman issues 50 bags of cement, he picks the cost code. He does not need to know about accounts or cost centres — the system takes those from the cost code and posts them correctly.

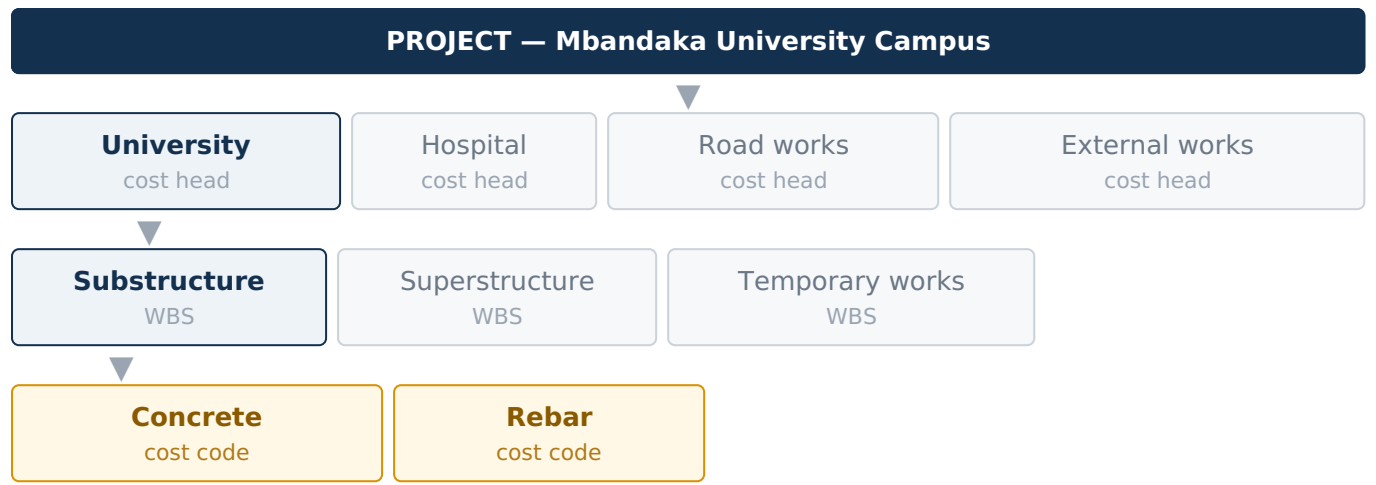
2. How work flows through the system

This is the whole journey, from agreeing a budget to paying for what was delivered. Each box is a screen someone fills in; the tags travel with the work all the way along.



3. How a job is broken down

Before anything can be tracked, the job has to be described. This is done once, at the start, and everything afterwards refers back to it.



Read it downwards

The University package contains the substructure, which is built from concrete and rebar. Costs roll back up the same way: every bag of cement issued against the Concrete code adds to the substructure, and to the University package, and to the project.

Only the highlighted path is shown

Each of the other packages has its own breakdown beneath it, and each part of the works its own cost codes. The tree is as deep as the job needs — a part of the works can have sub-parts.

Why the cost code matters most: it is the level at which a quantity surveyor actually works. The BOQ says you allowed 4,000 bags of cement for the substructure. The cost code is what lets the system tell you that 1,450 have gone out and you are 36% through the allowance.

4. What you call it, what the system calls it

The screens use standard ERPNext names in some places and construction names in others. This table maps them, so nobody gets lost.

What you say	On screen	What it does
Package	Cost Head	A major division of the job — the university block, the hospital, the roads.
Breakdown of works	WBS	The parts each package is built from. Sits in a tree, so parts can have sub-parts.
Cost code	Cost Code	A type of cost within the works. Carries the account and cost centre it should post to.
Bill of quantities	BOQ	Quantities and rates, approved through a sign-off process before anything is bought.
Budget allocation	WBS Allocation	Spreads the approved BOQ across the parts of the works, so each has its own budget.
Buying schedule	Procurement Plan	Works back from the date material is needed on site, using supplier lead times.
Requisition / PR	Material Request	Site asking for material, tagged to the cost code it is for.
Purchase order	Purchase Order	The commitment to a supplier. From this point the money is spoken for.
Shipment tracker	Shipment Tracking	Where an import is: sailed, arrived, in customs, on the road. Holds the documents too.
GRN	Purchase Receipt	What actually arrived. A receipt taken straight to site is flagged as an SMRN.
Landed cost	Landed Cost Voucher	Adds freight, duty and clearing onto the item so stock is valued at true cost.

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4. What you call it, what the system calls it — continued

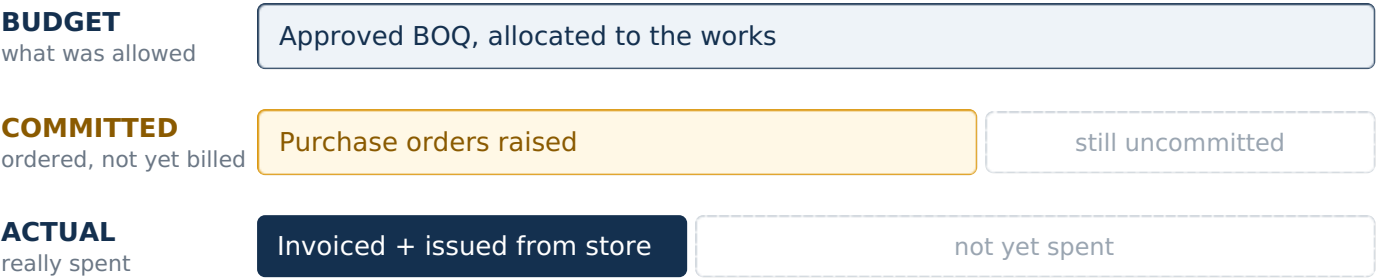
What you say	On screen	What it does
MIN — issue note	Stock Entry	Material leaving store for the job. This is the moment it becomes a cost.
MRN — receipt note	Stock Entry	Material arriving at the site store, usually after a transit leg.
Site return	Stock Entry	Surplus material coming back off site into the store, so it is not counted as used.
Work certificate	Work Certificate	Subcontractor work measured for the month, with retention withheld automatically.
Tool register	Tool Issue	Who is holding which tool. Moves the tool into that person's custody store.
Plant register	Asset	Machines, with make, model, capacity, warranty and their own numbering per category.
Breakdown record	Asset Repair	What failed, what it cost, which spares were used and how long the machine was down.
Spares list	Asset Spare Part	Which parts fit which machine, with minimum stock and preferred supplier.
Supplier bill	Purchase Invoice	What the supplier is charging, matched back to the order and the receipt.
Progress claim	Sales Invoice	What is billed to the client for work done, against the project.

One thing worth explaining to users: a material issue, a material receipt and a site return are all a *Stock Entry* on screen — the difference is the type chosen at the top. The numbering tells them apart: issues are **MIN-**, receipts are **MRN-**.

Why some names look unfamiliar: the system is built on ERPNext, which is used by every kind of business, so a few screens carry general names like “Stock Entry” or “Material Request”. The construction-specific parts — BOQ, WBS, Cost Head, Cost Code, Work Certificate, Tool Issue, Shipment Tracking — were built for you and carry the names you already use.

5. Where the money is tracked

This is the part that pays for the system. A cost passes through three states, and the system keeps them separate.



The gap between the middle bar and the bottom bar is the one that catches people out

On spend alone this package looks comfortable. But orders are already placed well beyond that, and those invoices are coming. The system shows both, so the overspend is visible months before it lands.

The reports that answer it

Report	What it tells you
Cost Code Wise Costing	Budget, committed, actual and balance for every cost code, WBS or package.
Budget vs Commitment	How much of each budget is already spoken for on purchase orders.
Cost to Complete	What each cost code will finish at, and whether that is above or below budget.
BOQ vs Consumption	Quantities allowed against quantities issued, flagging over-consumption.
Cash Flow Forecast	Money due in and out over the coming weeks, from commitments already made.
Working Capital	Receivables, payables, stock, work in progress and retention, month by month.

6. The construction-specific parts

Standard ERP handles buying, stock and accounts. These are the pieces that were added because a construction business needs them and off-the-shelf systems do not have them.

Imports tracked door to door

A single record follows a container from the supplier's factory to the site store: FAT inspection, stuffing, sailing, arrival, customs, inland haulage. It holds the bill of lading, certificate of origin, duty receipt and delivery order, and records demurrage and detention when a container is held too long. Reports show turnaround by stage, so you can see whether the delays are at the port, at customs, or on the road.

Retention handled properly

When subcontractor work is certified, the system withholds the retention percentage and posts it to a separate **Retention Payable** account. The subcontractor is paid the net amount; the retention stays on the balance sheet until the defect liability period expires, at which point it is released deliberately rather than forgotten.

Small tools tracked by person

Tools are not treated as fixed assets — they are stock, issued into a custody store in an individual's name. At any point you can list what each person is holding, how long they have had it, and what it is worth. Lost tools are written off with the amount recovered from the holder recorded against them.

Labour costed to the works

Attendance is recorded against a project and site, and timesheets against a WBS and cost code. Staff who fill timesheets have their hours costed directly; site labour who do not have their payroll spread across the projects they were present on, in proportion to days worked. Every figure says which basis it used, so nothing is silently half-measured.

Plant that earns or costs

Each machine is numbered by its own category — HEQ for heavy equipment, VEH for vehicles — and carries its make, model, capacity, serial and warranty. Breakdowns, repair costs and spare parts are recorded against it, and utilisation is measured against downtime, so the decision to keep a machine or hire one is based on its own numbers.

7. What you see when you log in

Everything sits under one heading in the menu, **A3 Constructa**, with nine areas beneath it. Each area is a page of shortcuts to the screens and reports for that part of the business.

Area	What it covers
Master Data	The job itself: packages, breakdown of works, cost codes, items, suppliers, stores, ports and approval rules.
Planning & Budgeting	The BOQ, its approval, the budget spread across the works, and the buying schedule.
Procurement	Requisitions, quotations, vendor selection, purchase orders and subcontract certificates.
Procurement Classification	How materials, services, spares, tools and rentals are grouped for reporting.
Delivery & Logistics	Shipment tracking, goods receipts, landed cost and the document trail.
Inventory Movement	Stock in the stores, issues to the job, transit legs and site returns.
Asset & Equipment	Plant register, maintenance, spares and the small tools register.
HR & Time	People, attendance, shifts, leave, payroll, expenses and timesheets.
Finance & Accounting	Ledger, payables, receivables, banking, tax, cost control and the finance dashboard.

The finance dashboard

The Finance area opens with the figures a director asks for first: total receivable and payable, what is overdue, cash and bank balance, retention held, how much is committed but not yet billed, and budget utilisation. Underneath are trend charts for cash flow, ageing, budget variance and working capital.

A note on permissions: the system knows who is looking. A project manager can be limited to their own project's figures while finance sees the company-wide position, using the same screens.

8. A worked example

The demonstration system is loaded with one real-shaped project so you can follow a single cost from beginning to end. Here is the cement.

Stage	What happens	Result
1. Allowed	The BOQ allows 4,000 bags for the substructure	Budget set against the Concrete cost code
2. Planned	Needed on site in two weeks; supplier takes three	The system says the order should already have been placed
3. Ordered	2,000 bags ordered from the cement works	Money now shows as <i>committed</i> , not yet spent
4. Delivered	Goods receipt into the central store	Stock on hand rises; the order stops being a commitment
5. Moved	600 bags sent upriver by barge; 450 arrive so far	150 bags visible as stock in transit
6. Used	1,450 bags issued to the substructure	Now a real cost against the Concrete cost code
7. Billed	Supplier invoice received, due date passed	Shows in payables ageing and the overdue figure
8. Reported	Cost Code Wise Costing is opened	Budget, committed, actual and balance for that one cost code

The same trail exists for the imported steel, the blockwork subcontract with its retention, the excavator and its breakdown, the tools in people's custody, and the two progress claims to the client — one paid, one overdue.

Worth showing live: open **Cost Code Wise Costing**, then change the grouping from Cost Code to WBS to Package. It is the same money answered three ways, and it makes the point about the tags better than any explanation.

What the demonstration does not show

Two screens are deliberately empty: subcontract purchase orders, which need the standard subcontracting setup configured, and internal transfer despatch notes, which need an internal customer defined. Both are configuration decisions for the live system rather than missing functionality.